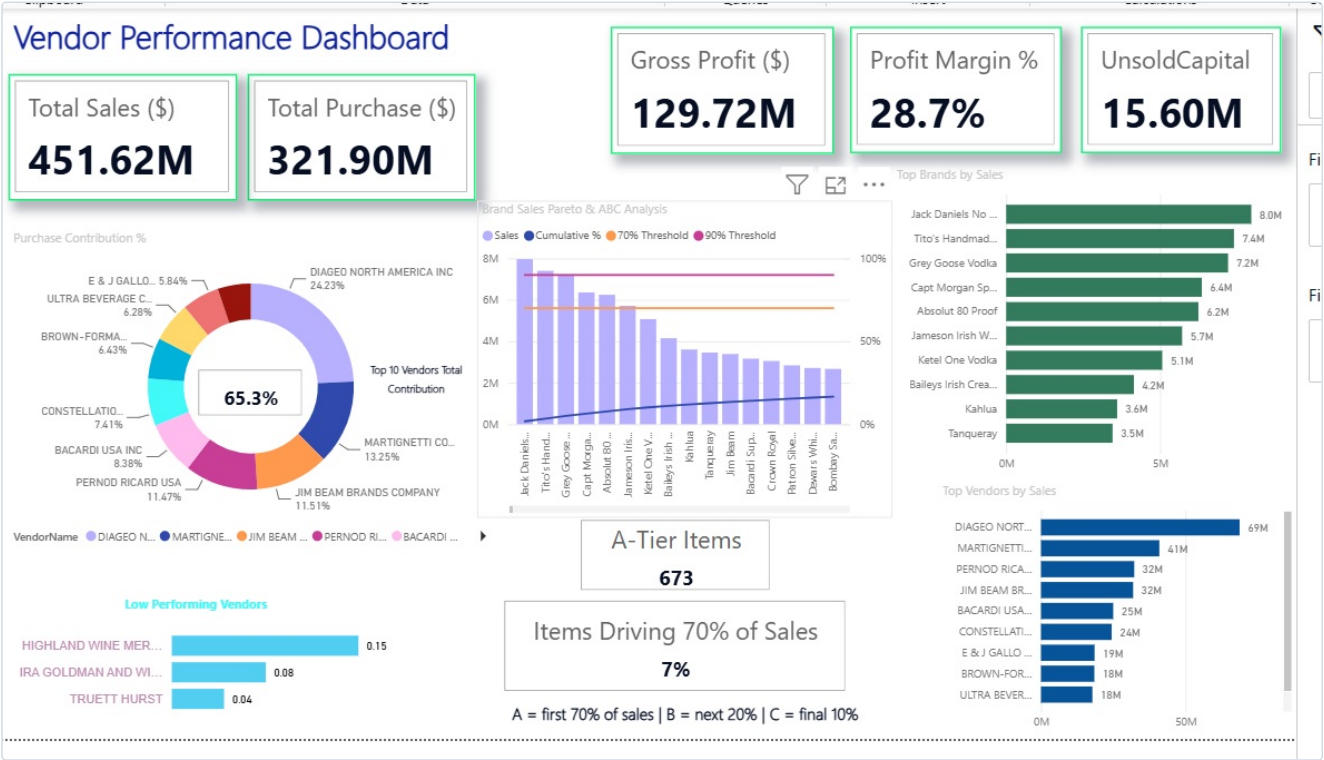


VENDOR PERFORMANCE ANALYSIS

End-to-End Data Analytics Project
SQL + Python + Statistical Analysis + Power BI



Portfolio Flagship Project

A business-focused analysis of purchasing, sales, profitability, vendor concentration and inventory efficiency.

Prepared as the master findings document for GitHub, LinkedIn, resume and interview use.

1. Executive Summary

This project evaluates vendor and product performance across purchasing, sales, gross profit, profit margin, inventory turnover and unsold capital. The objective is to identify where procurement concentration, product concentration, pricing opportunities and inventory inefficiencies can be converted into actionable business decisions.

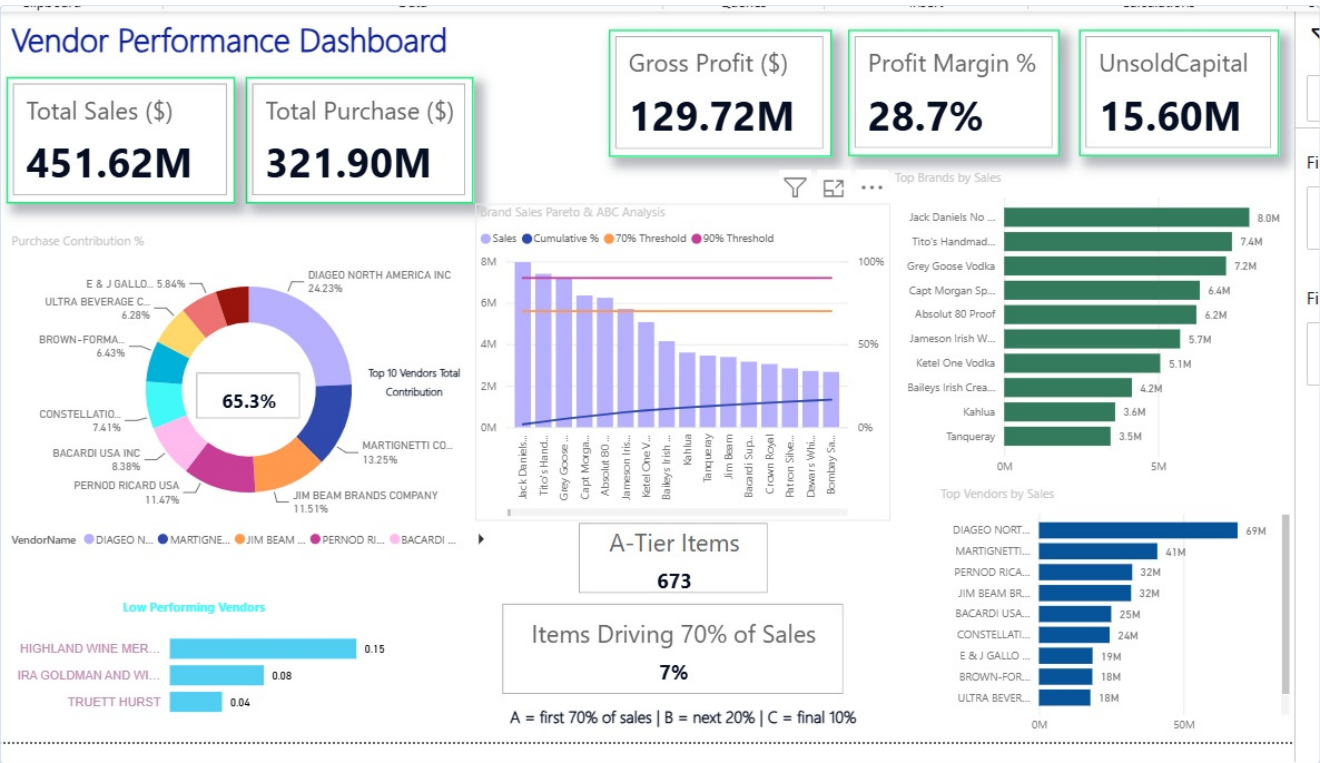
The final Power BI dashboard reports **\$451.62M** in total sales, **\$321.90M** in total purchases, **\$129.72M** in gross profit, a **28.7%** profit margin and **\$15.60M** in unsold capital. The dashboard also identifies **673 A-tier items** and shows that only **7%** of items account for the first 70% of sales.

Headline finding: performance is highly concentrated. A small group of vendors and products drives a large share of commercial activity, while a separate group of slow-moving and low-sales products ties up capital or requires targeted commercial intervention.

Two additional findings emerged from deeper analysis of fields not covered by the base workflow: vendors pursue fundamentally different strategies around catalog breadth versus revenue concentration (Diageo North America generates **5.9x** more revenue per SKU than Martignetti Companies, the portfolio's highest-SKU-count vendor), and freight cost runs a consistent **~0.5%** of purchase value across nearly the entire vendor base — except two vendors whose logistics cost is **90-100x** higher than typical, a concrete red flag for procurement follow-up.

The project therefore recommends a differentiated strategy: protect availability of high-impact A-tier products, strengthen supplier-risk management, use bulk purchasing selectively where demand supports it, actively manage slow-moving inventory, investigate the two freight-cost outliers, and evaluate vendors using multiple dimensions rather than sales or margin alone.

Final dashboard output



2. Business Problem & Analytical Framework

Business questions addressed

- Which vendors contribute the most to purchasing and sales?
- How concentrated is supplier dependency?
- Which products drive the majority of sales?
- Which vendors and products are underperforming?
- Where is capital tied up in unsold inventory?
- Does bulk purchasing reduce unit purchase cost?
- Are high-margin products necessarily high-performing products?
- What relationships exist between price, volume, turnover and profitability?

Analytical workflow

The work follows an end-to-end analytics flow: data preparation → exploratory analysis → metric engineering → correlation/outlier analysis → vendor/product segmentation → statistical validation → Power BI dashboarding → business recommendations.

Core metrics

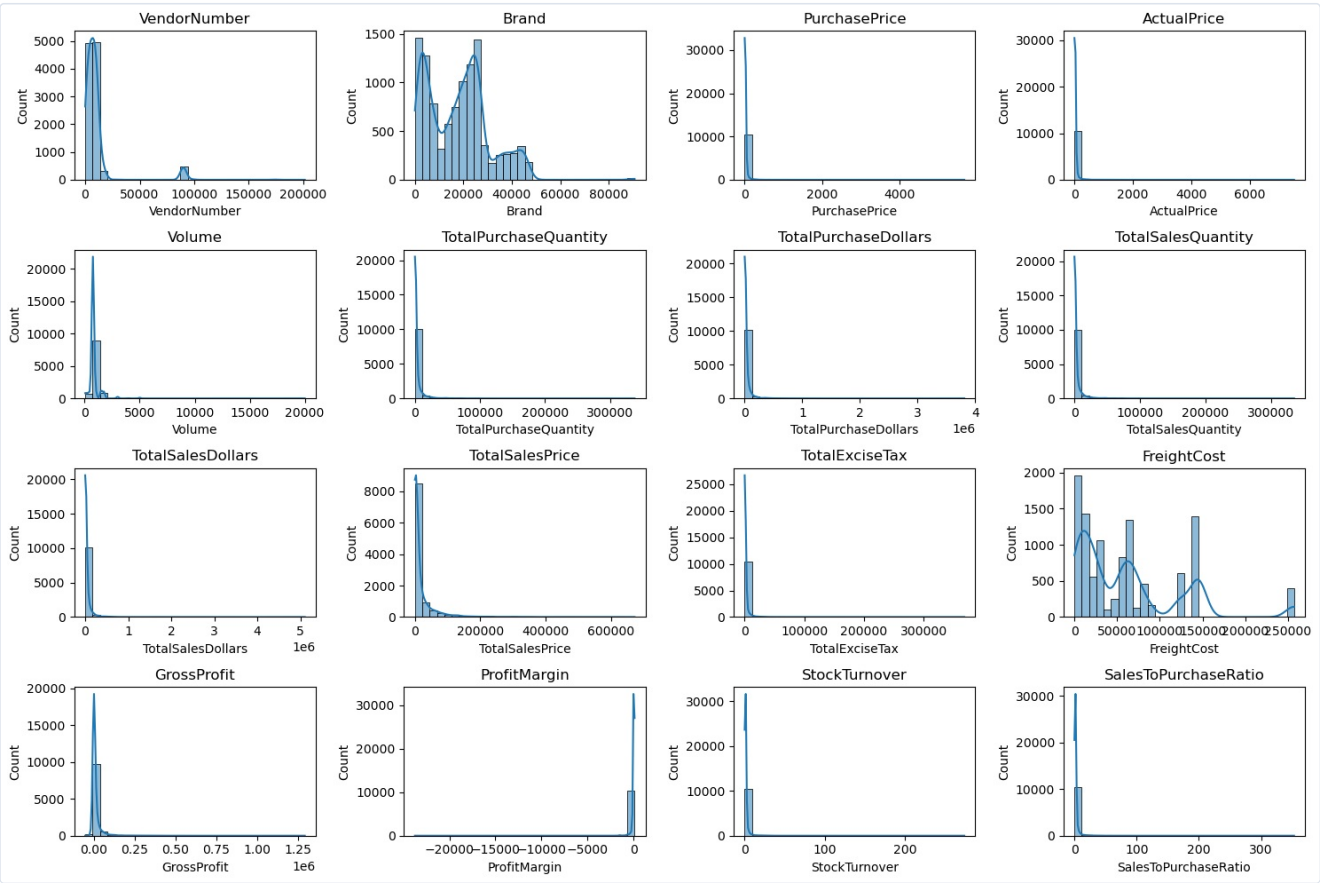
Metric	Business meaning
Total Sales	Value of sales generated by the analyzed portfolio
Total Purchase	Value of procurement / purchase activity
Gross Profit	Sales value less purchase cost in the project model
Profit Margin	Gross profit expressed relative to sales
Stock Turnover	Indicator of how quickly inventory is moving
Unsold Capital	Capital tied up in inventory not yet sold
ABC Classification	Segmentation based on cumulative sales contribution

3. Exploratory Data Analysis

The exploratory analysis was used to understand the structure and quality of the numerical fields before drawing business conclusions. The analysis examined distributions, central tendency, spread, minimum/maximum values, zero values and extreme observations across all 10,692 records.

Field	Mean	Std Dev	Min	25%	Median	75%	Max
PurchasePrice	24.39	109.27	0.36	6.84	10.46	19.48	5,681.81
ActualPrice	35.64	148.25	0.49	10.99	15.99	28.99	7,499.99
TotalSalesDollars	42,239.07	167,655.27	0.00	729.22	5,298.05	28,396.92	5,101,920
GrossProfit	12,132.38	46,224.34	-52,002.78	52.92	1,399.64	8,660.20	1,290,668
ProfitMargin (%)	-∞	—	-∞	13.32	30.41	39.96	99.97
StockTurnover	1.71	6.02	0.00	0.81	0.98	1.04	274.50

The summary statistics show substantial variation across purchase price, actual price, quantities, sales dollars, freight cost and gross profit. Several fields are strongly right-skewed, meaning a relatively small number of observations have very large values compared with the majority.



Distribution plots for all numerical columns — generated from the project's own Jupyter notebook

Distribution outputs indicate that many monetary and quantity variables are concentrated at lower values with long right tails. This supports the need to examine outliers and avoid relying only on averages.

4. Data Quality, Negative Values & Outliers

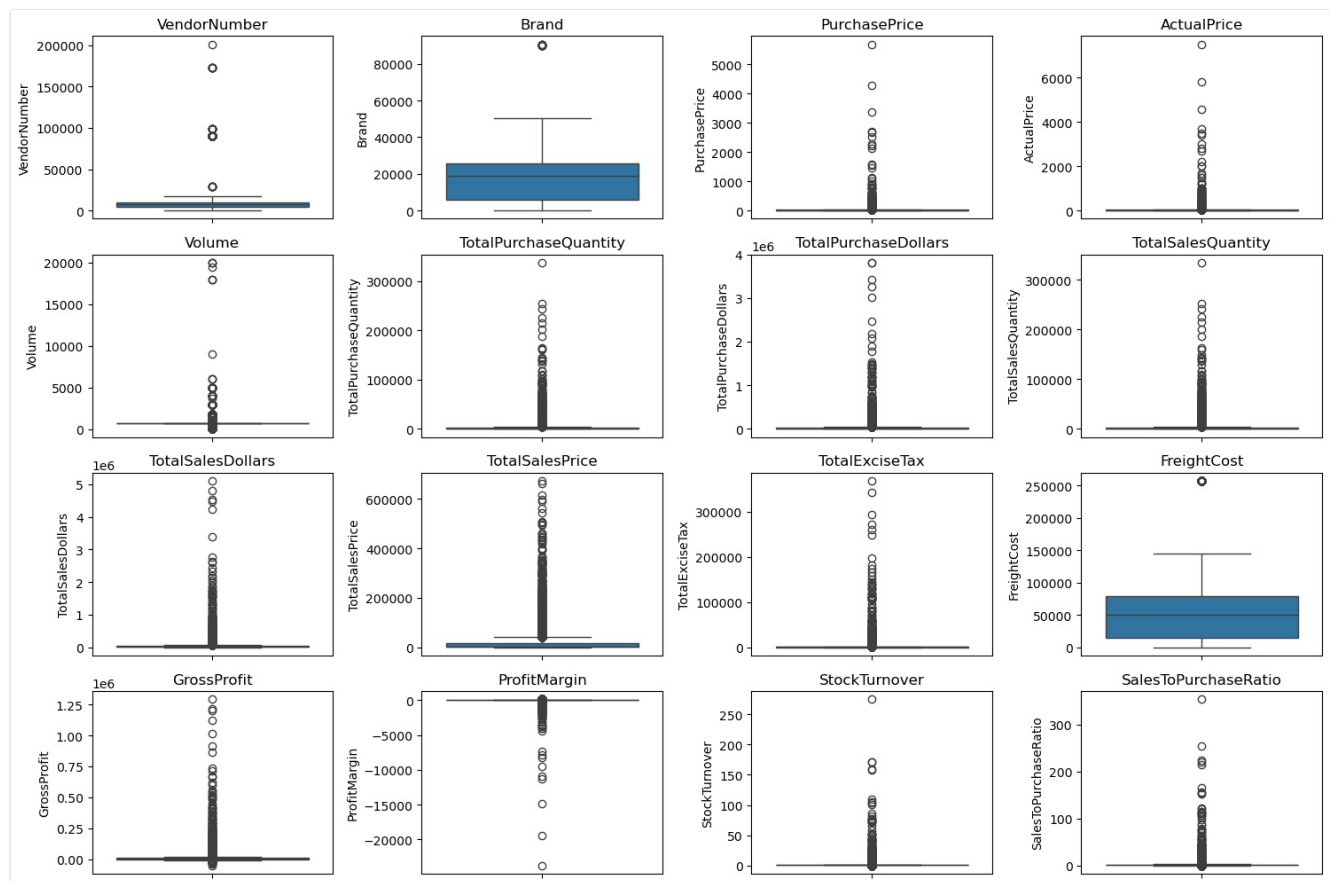
The analysis explicitly investigated negative and zero values before building business conclusions. Gross profit contains negative observations, while some products have zero sales quantity and zero sales dollars.

Negative & Zero Values

Gross Profit: Minimum of -52,002.78, indicating potential losses due to high costs or heavy discounts. This could be due to selling products at lower prices than their purchase costs.

Profit Margin: Has a minimum of $-\infty$, which suggests instances where revenue is zero or even lower than the total cost, leading to extreme negative profit margins.

Total Sales Quantity & Sales Dollars: Some products show zero sales, indicating they were purchased but never sold. These may be slow-moving or obsolete stock, leading to inventory inefficiencies.



Outlier detection via boxplots across all numerical columns

Outliers Detected by High Standard Deviations

Purchase & Actual Prices: The maximum values (5,681.81 & 7,499.99) are significantly higher than the mean (24.39 & 35.64), indicating premium product offerings.

These observations can represent genuine premium products, bulk transactions or unusual logistics costs, but they should be interpreted carefully rather than automatically removed.

5. Overall Financial Performance

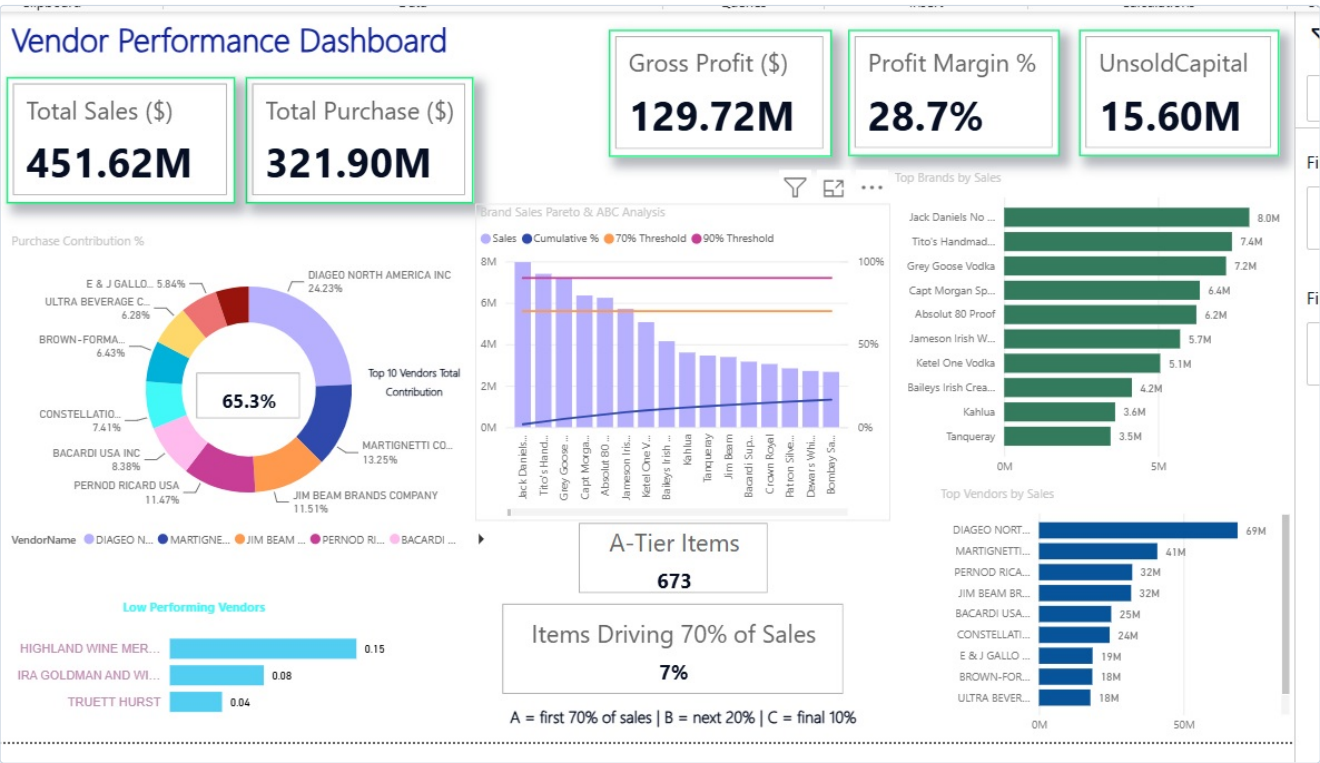
Total Sales — \$451.62M: The analyzed portfolio generates substantial sales value across the vendor/product base.

Total Purchase — \$321.90M: Procurement expenditure represents the underlying cost base in the project model.

Gross Profit — \$129.72M: The dashboard shows a large absolute gross-profit contribution.

Profit Margin — 28.7%: Overall gross profitability is healthy, but the later vendor-level analysis shows that margin is distributed unevenly.

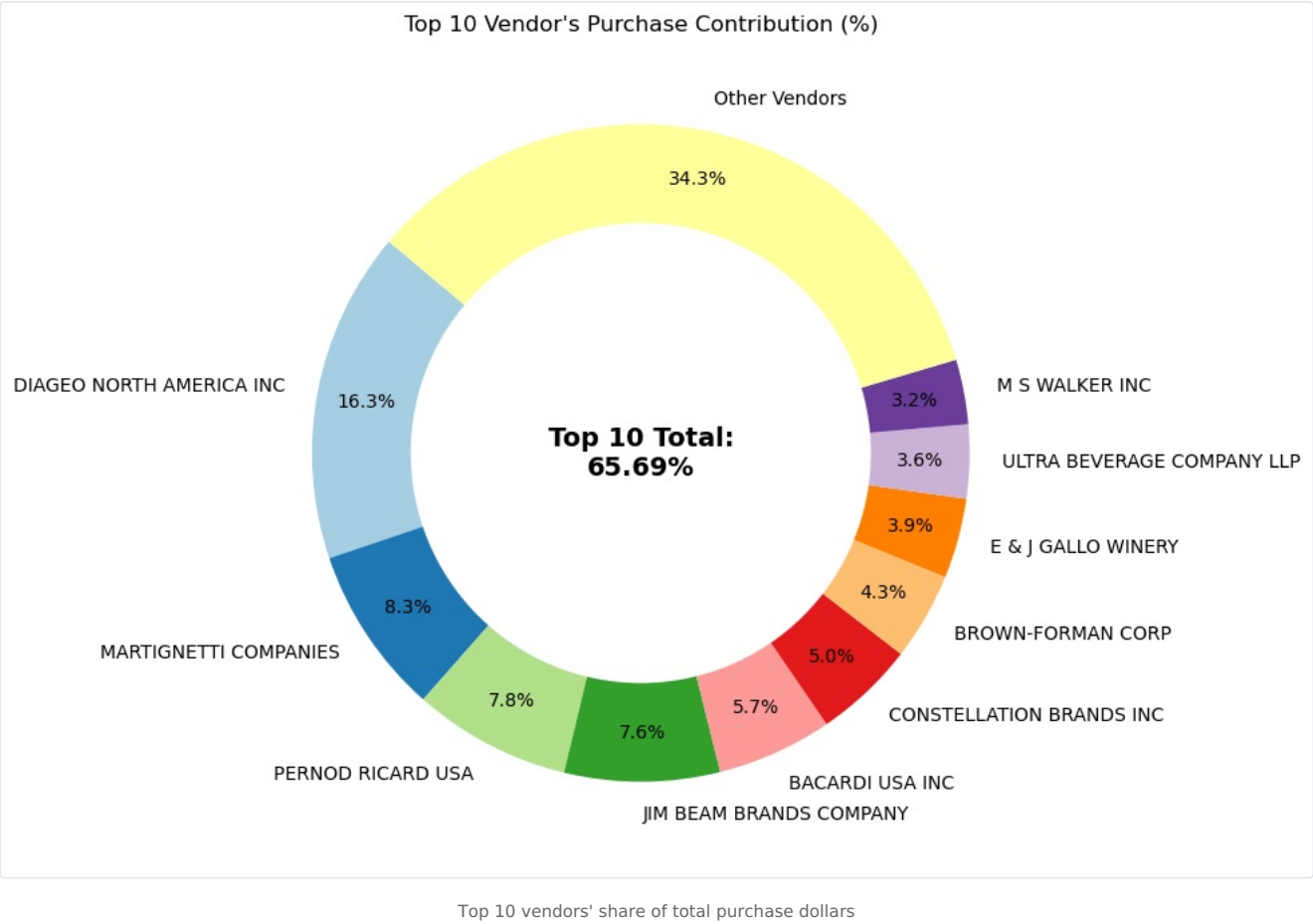
Unsold Capital — \$15.60M: A significant amount of capital is tied up in unsold inventory and therefore deserves dedicated working-capital attention.

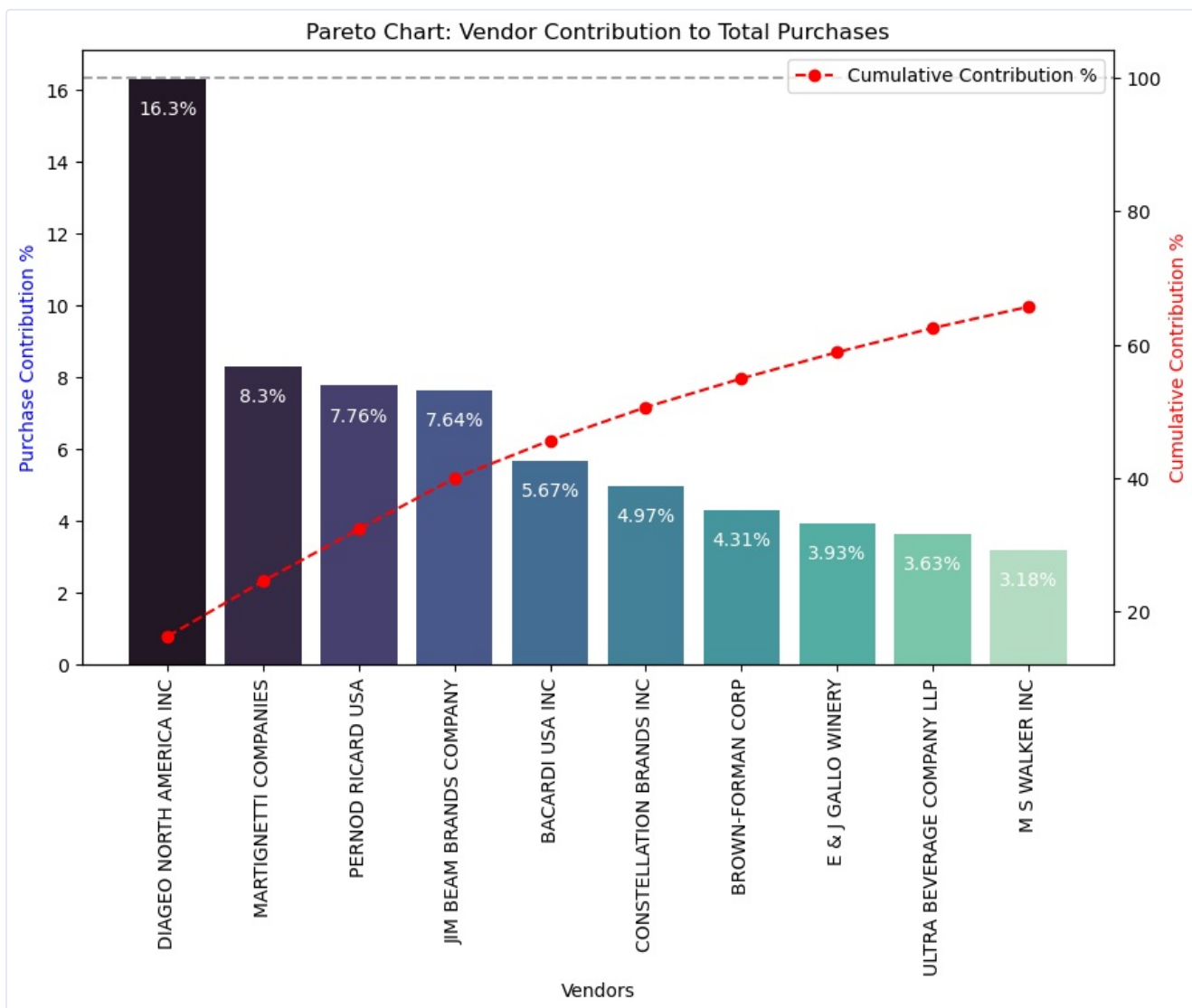


Management takeaway: the portfolio is profitable overall, but improving performance is less about simply increasing total sales and more about optimizing the mix of vendors, products, inventory and margin.

6. Vendor Concentration & Supplier Risk

The vendor contribution analysis shows that the **top 10 vendors account for 65.69% of total purchases**. The concentration indicates that procurement activity is heavily weighted toward a relatively small supplier group.





Pareto view of vendor purchase contribution with cumulative %

Top vendors by sales include Diageo North America (\$68.0M), Martignetti (\$39.3M), Pernod Ricard USA (\$32.1M), Jim Beam Brands (\$31.4M) and Bacardi USA (\$24.9M).

Business implication: concentration can be beneficial when it creates scale, negotiating leverage and reliable supply, but it also increases exposure if a major supplier faces price increases, allocation constraints, quality problems or disruption.

Recommendation: maintain strategic relationships with high-performing suppliers while qualifying alternative sources for critical categories. Diversification should be risk-based rather than a blanket reduction in purchases from top vendors.

7. ABC Analysis & Product Concentration

The dashboard uses an ABC framework in which A represents the first 70% of sales, B the next 20%, and C the final 10%. The analysis identifies **673 A-tier items** and shows that only **7%** of items drive the first 70% of sales.



A = first 70% of sales | B = next 20% | C = final 10%

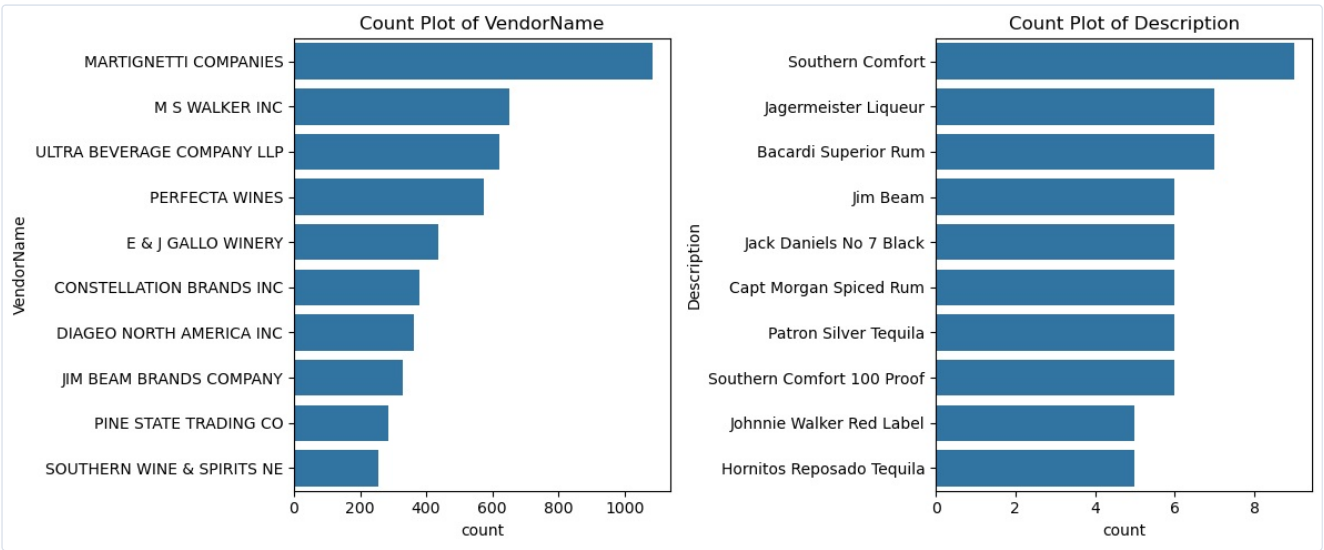
This is a critical inventory-management finding. The portfolio should not be managed uniformly because a small number of products have disproportionate commercial importance.

Tier	Policy
A-tier	Frequent monitoring, stronger replenishment discipline, demand forecasting and high service-level attention.
B-tier	Balanced monitoring based on demand and margin.
C-tier	Lower purchasing priority, tighter minimum/maximum controls and review for clearance or delisting where appropriate.

Key takeaway: inventory resources should follow business impact rather than SKU count.

8. Portfolio Breadth vs. Revenue Concentration

Beyond how much each vendor sells, it's worth asking how they sell it — through a wide catalog of many products, or a concentrated set of flagship brands. Counting distinct product line-items per vendor surfaces a pattern the sales-ranking view alone does not show.



Vendors and brands with the highest number of distinct line-items in the portfolio

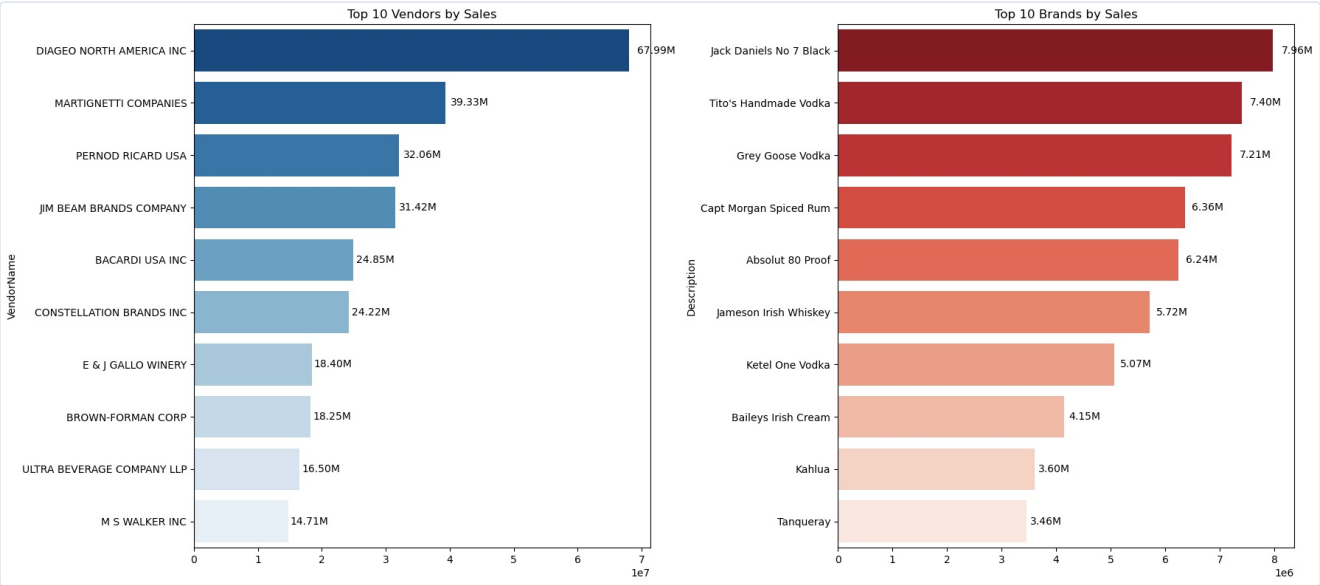
Vendor	SKU Count	Total Sales	Revenue per SKU
Diageo North America Inc	396	\$68.74M	\$173,585
Martignetti Companies	1,388	\$40.96M	\$29,510

Martignetti Companies carries by far the broadest catalog in the portfolio — 1,388 distinct product line-items, more than 3.5× the next-highest vendor. Yet it ranks only **#2** by total sales, behind **Diageo North America**, which sells through just 396 line-items.

Key insight: Diageo generates roughly **5.9× more revenue per SKU** than Martignetti. This reflects two distinct and equally valid vendor strategies — concentrated flagship-brand power versus broad catalog coverage — and suggests vendor scorecards should weigh *revenue efficiency per line-item* alongside total sales, not sales volume alone. A vendor with fewer, higher-performing SKUs may warrant a different negotiating and replenishment strategy than one whose value comes from breadth.

9. High-Sales Brands & Commercial Opportunities

The dashboard highlights the leading brands and vendors by sales. Jack Daniels No 7 Black leads at approximately \$7.96M, followed by Tito's Handmade Vodka at \$7.40M and Grey Goose Vodka at \$7.21M. Other leading products include Captain Morgan Spiced Rum, Absolut 80 Proof, Jameson Irish Whiskey and Ketel One Vodka.



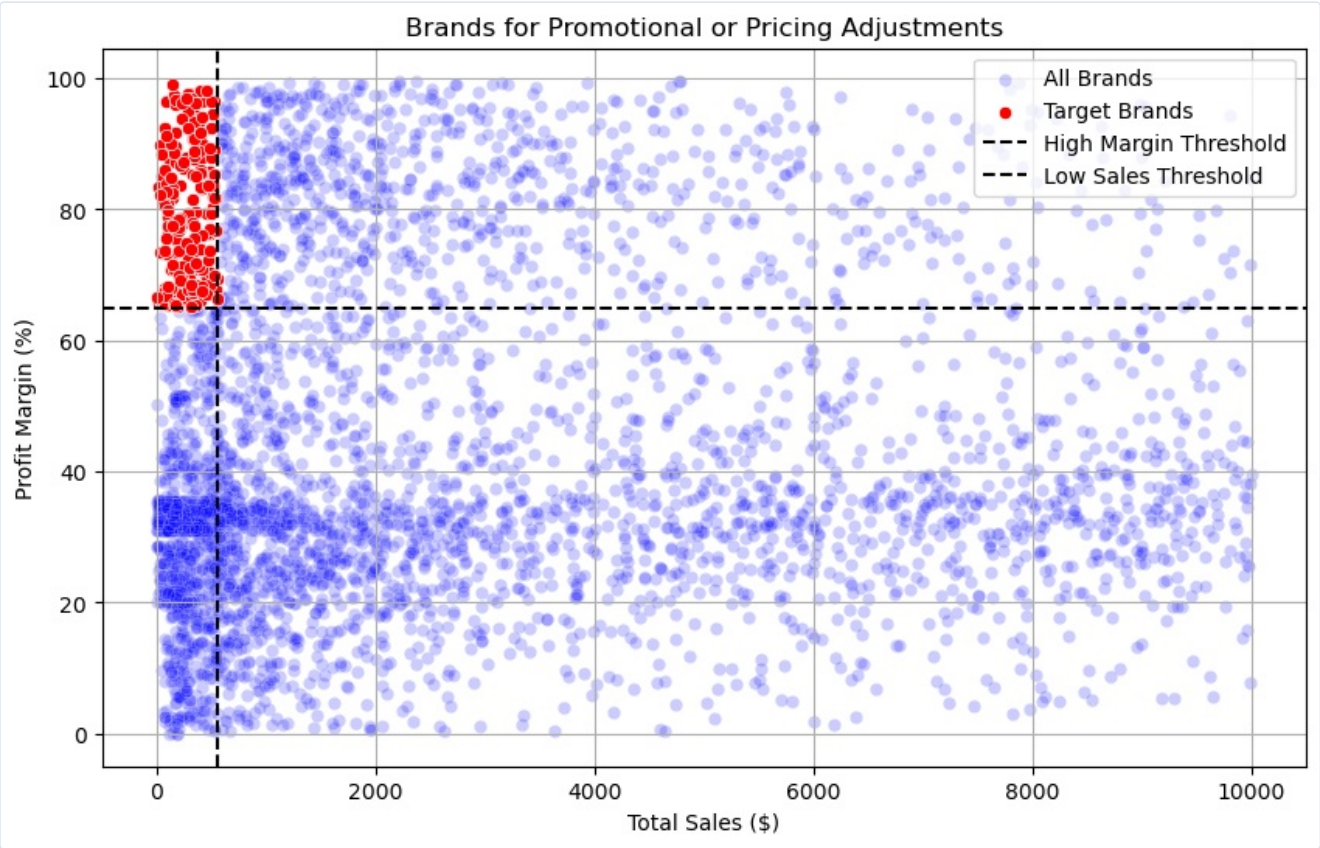
Top 10 vendors and top 10 brands by total sales dollars

High-sales products should receive priority in replenishment, supplier coordination and stockout prevention because disruptions in these products can have a disproportionate effect on revenue.

Commercial principle: protect availability first for high-impact products, then optimize margin and purchasing terms around the demand that already exists.

10. Low-Sales / High-Margin Brands

The Python analysis identified **198 brands** exhibiting lower sales while maintaining relatively high profit margins — the bottom 15% by sales combined with the top 15% by profit margin. Examples include Santa Rita Organic, Debauchery, Concannon Glen Ellen, Crown Royal Apple, Sauza, Nanbu Bijin and Dad's Hat Rye Whiskey.



All brands (blue) vs. target brands meeting the low-sales / high-margin criteria (red)

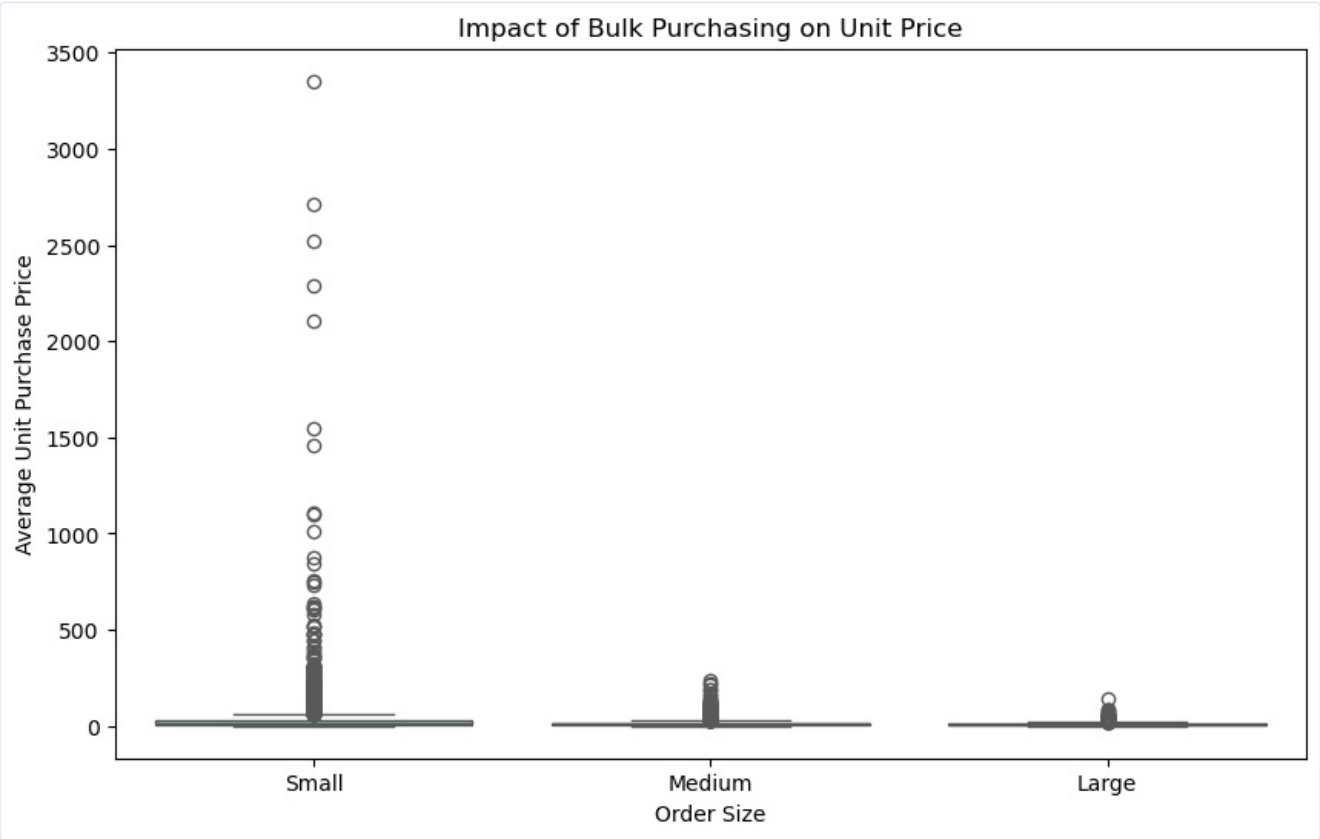
Brand	Total Sales (\$)	Profit Margin
Santa Rita Organic Svgn Bl	9.99	66.47%
Debauchery Pnt Nr	11.58	65.98%
Concannon Glen Ellen Wh Zin	15.95	83.45%
Crown Royal Apple	27.86	89.81%
Sauza Sprklg Wild Berry Marg	27.96	82.15%
Nanbu Bijin Southern Beauty	535.68	76.75%
Dad's Hat Rye Whiskey	538.89	81.85%

These brands should not automatically be classified as failures. Their high margins suggest that the commercial problem may be demand, reach, distribution, price positioning or awareness rather than product economics.

Recommended tests: targeted promotions, improved distribution, shelf placement, bundles, selective price adjustments and customer-level targeting. The objective is to increase volume without unnecessarily destroying the margin advantage.

11. Bulk Purchasing & Unit-Cost Advantage

The analysis shows a strong relationship between order size and average unit purchase price.



Unit purchase price distribution by order size

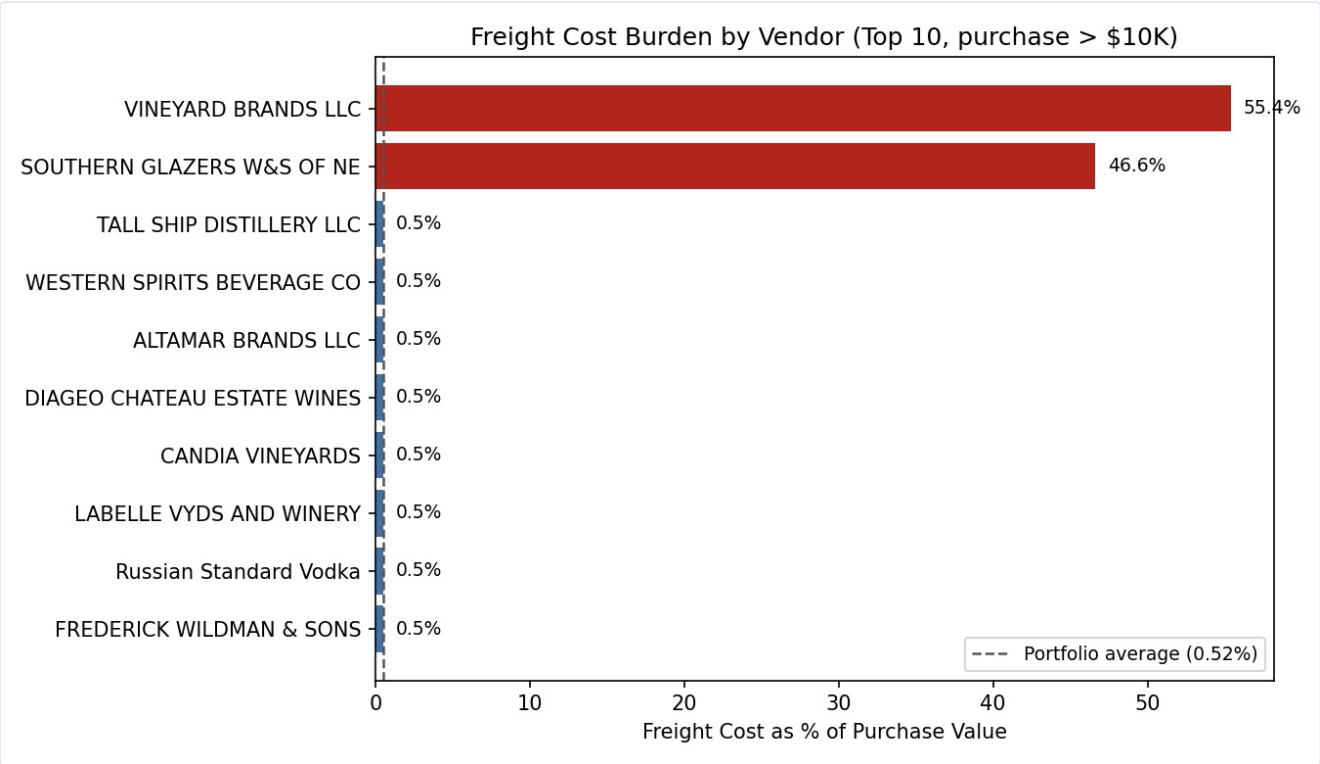
Order Size	Avg. Unit Purchase Price
Small	\$39.07
Medium	\$15.49
Large	\$10.78

Average unit purchase price falls from approximately \$39.07 for small orders to \$15.49 for medium orders and \$10.78 for large orders — a substantial unit-cost advantage for larger purchasing quantities.

Procurement implication: volume negotiation can create meaningful cost savings, but the decision must be constrained by demand and inventory turnover. Buying too much of a slow-moving product can convert a unit-cost saving into a working-capital problem. The strongest policy is therefore **selective bulk purchasing for products with proven demand and acceptable turnover.**

12. Freight Cost Efficiency & Logistics Risk

Freight cost is recorded once per vendor rather than per product line, so it was checked separately from the per-brand metrics above. Comparing each vendor's total freight cost against its total purchase value reveals how consistent — or not — logistics efficiency is across the supplier base.



Freight cost as a percentage of purchase value, top 10 vendors by ratio (vendors with purchase value > \$10K)

Across nearly the entire vendor base, freight cost holds remarkably steady at approximately **0.52% of purchase value** — a tight, consistent logistics cost structure. Two vendors break sharply from this pattern:

Vendor	Freight Cost	Purchase Value	Freight % of Purchase
Vineyard Brands LLC	\$6,070	\$10,952	55.4%
Southern Glaziers W&S of NE	\$19,017	\$40,804	46.6%

Key insight: these two vendors carry a freight burden **90-100x higher** than the portfolio norm. Since every other vendor sits tightly around 0.5%, this is unlikely to be a systemic data issue — it points to something specific to these two accounts, such as small/inefficient shipment sizes, a shipping-terms mismatch, or a data-entry anomaly on the invoice. This is a concrete, low-effort item worth a direct follow-up with procurement or logistics before the next order cycle.

13. Inventory Turnover & Unsold Capital

Inventory efficiency is a central theme of the project. The final Power BI dashboard reports **\$15.60M in unsold capital**. Slow-moving inventory can increase storage requirements, reduce cash-flow efficiency and prevent capital from being deployed into faster-moving products.

\$15.60M

Total Unsold Capital

Vendors with lowest inventory turnover

Vendor	Stock Turnover
Alisa Carr Beverages	0.615
Highland Wine Merchants LLC	0.708
Park Street Imports LLC	0.751
Circa Wines	0.756
Dunn Wine Brokers	0.766
Centeur Imports LLC	0.774

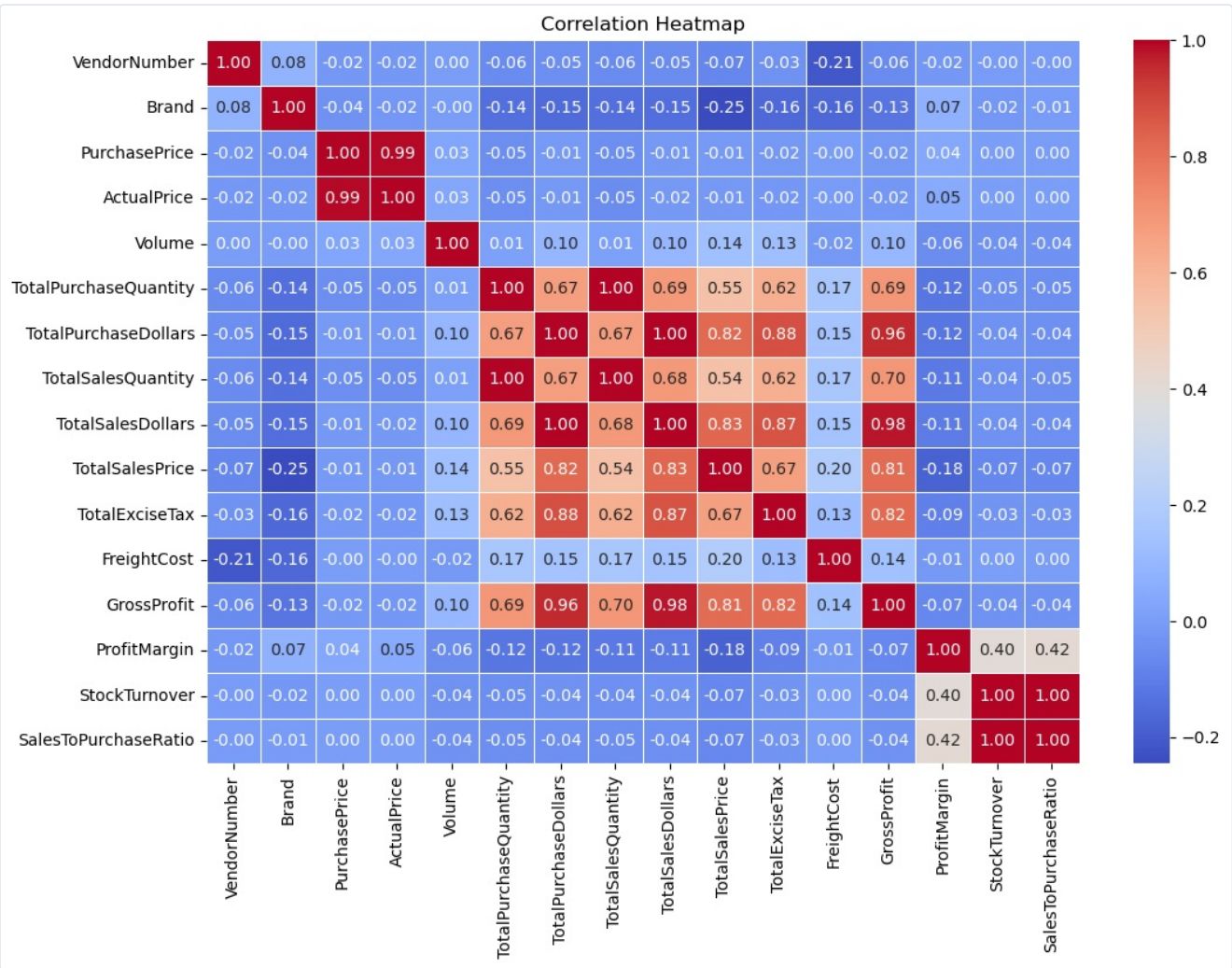
Vendors with largest capital tied up in unsold inventory

Vendor	Unsold Inventory Value
Diageo North America Inc	722.21K
Jim Beam Brands Company	554.67K
Pernod Ricard USA	470.63K
William Grant & Sons Inc	401.96K
E & J Gallo Winery	228.28K
Sazerac Co Inc	198.44K

Recommended inventory actions: prioritize high-value slow-moving stock, adjust purchase quantities, review minimum-order quantities, launch targeted clearance activity where justified, and revise storage/replenishment policies. The goal is not maximum turnover at any cost; it is the right balance between availability, margin and capital efficiency.

14. Correlation Analysis

The correlation matrix highlights several strong relationships among quantities, sales dollars and gross profit. Purchase quantity and sales quantity are strongly correlated in the analyzed data, while sales/purchase dollar measures and gross profit are also strongly related.



Correlation heatmap across all numerical fields

Purchase Price vs. Total Sales Dollars & Gross Profit: Weak correlation (-0.01 and -0.02), indicating that price variations alone do not significantly impact sales revenue or profit.

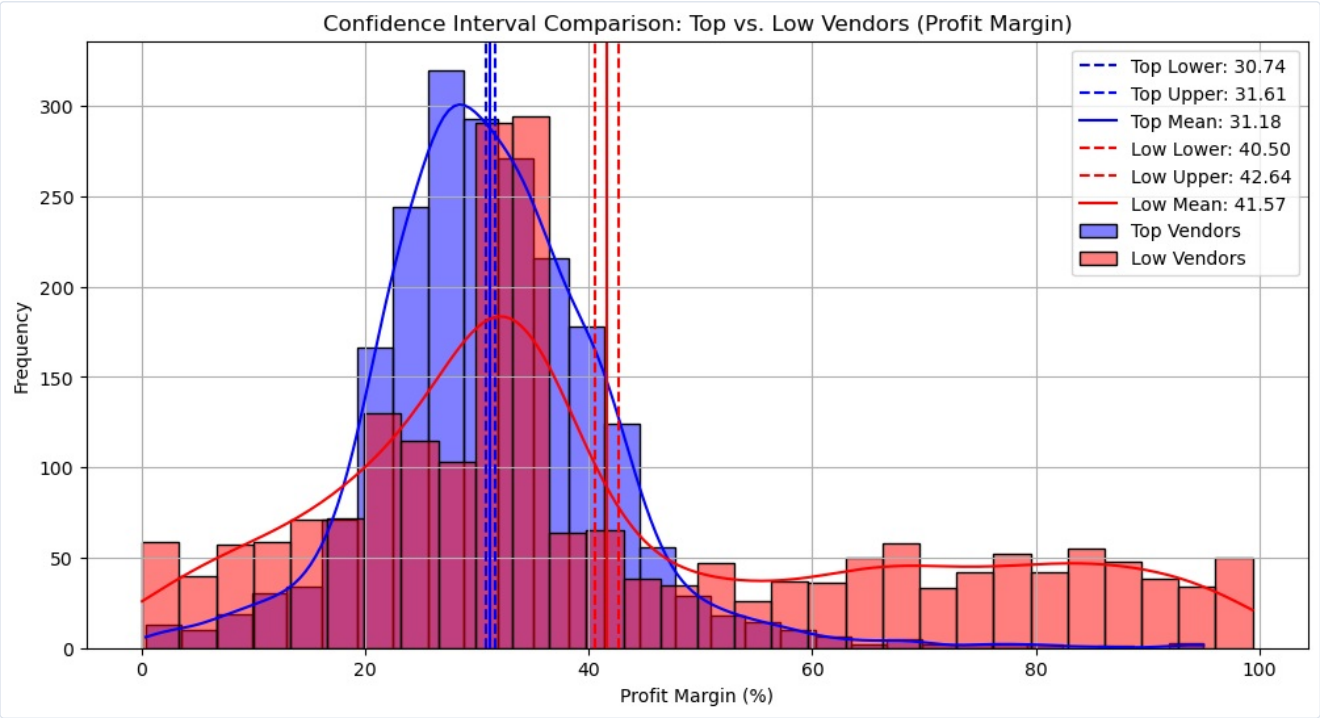
Total Purchase Dollars vs. Gross Profit: Strong correlation (0.96), reflecting that gross profit scales closely with purchase volume in this business.

Profit Margin vs. Stock Turnover / Sales-to-Purchase Ratio: Moderate positive correlation (0.40 and 0.42), suggesting faster-moving, better-priced products also tend to carry somewhat healthier margins.

Important interpretation: purchase price alone has a very weak relationship with total sales dollars and gross profit in the analysis. This means procurement optimization cannot be reduced to negotiating a lower unit price; volume, demand, mix and selling economics also matter.

15. Profit Margin Comparison: Top vs. Low-Performing Vendors

Vendors were split by total sales into a top-performing group (≥ 75 th percentile) and a low-performing group (≤ 25 th percentile), then compared on profit margin.



95% confidence interval comparison of profit margin: top vs. low-performing vendors

31.18% Top Vendors' Mean Margin (95% CI: 30.74%–31.61%)	41.57% Low Vendors' Mean Margin (95% CI: 40.50%–42.64%)
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Low-performing vendors maintain higher margins but struggle with sales volumes — potentially due to premium pricing, lower operational overhead, or serving niche markets rather than mass distribution.

Key insight: low-performing vendors can have higher margins than top-selling vendors. This demonstrates why vendor performance must combine sales contribution, profitability and inventory behavior instead of relying on a single KPI.

For top-performing vendors: consider selective price adjustments, cost optimization, or product bundling to lift margin without materially affecting sales volume.

For low-performing vendors: the margin advantage is already there — the constraint is reach. Broader distribution, targeted marketing, or competitive pricing on select lines could grow revenue while largely preserving the existing margin profile.

16. Statistical Validation

The project formulated a hypothesis test to determine whether the difference in profit margins between the vendor groups was statistically meaningful.

Hypothesis	Statement
H ₀ (Null)	No significant difference in profit margins between top and low-performing vendors.
H ₁ (Alternative)	A significant difference exists in profit margins between the two vendor groups.

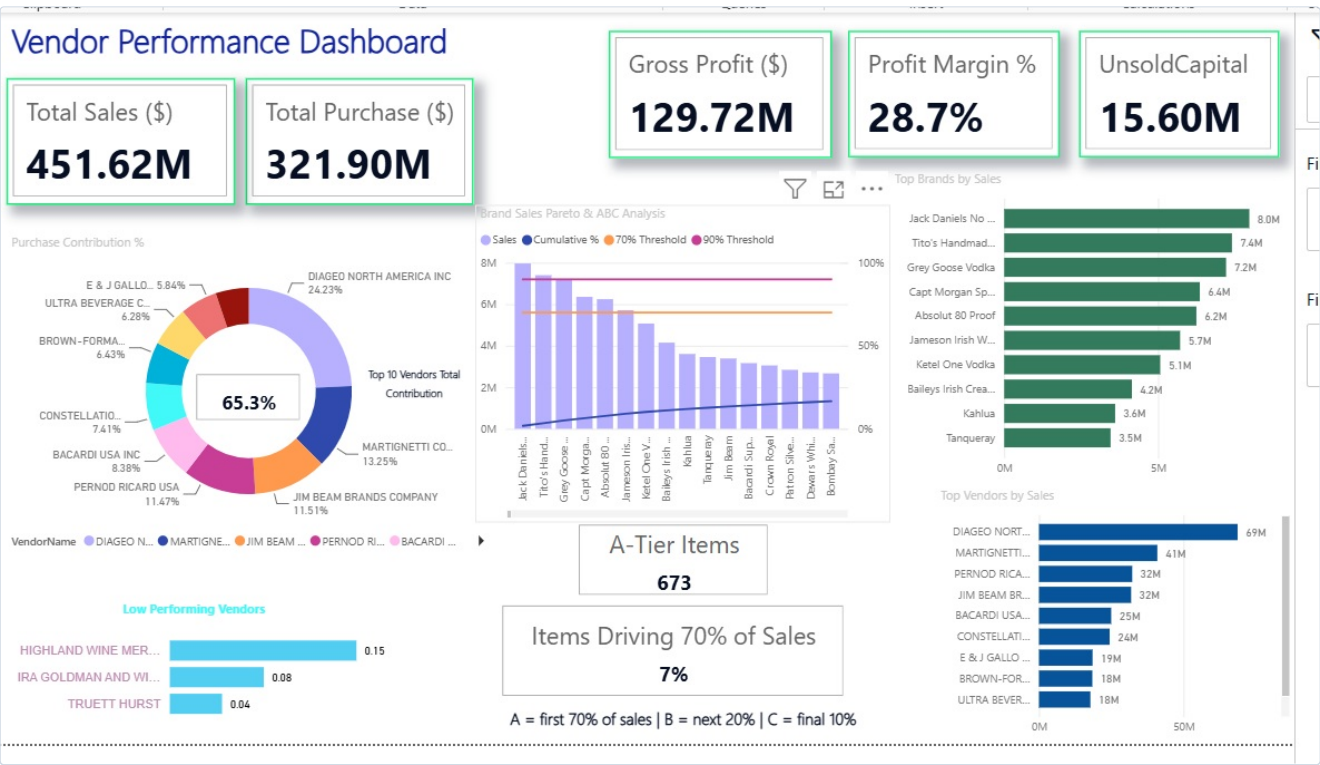
-17.67 T-Statistic	<0.0001 P-Value
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Result: the null hypothesis is rejected, confirming that the two groups operate under distinctly different profitability models.

Implication: high-margin vendors may benefit from better pricing strategies, while top-selling vendors could focus on cost efficiency. Vendor strategies should be differentiated — high-margin/low-volume vendors may require commercial growth initiatives, while high-sales vendors may offer greater opportunities for cost and margin optimization.

17. Final Power BI Dashboard

The final dashboard consolidates the analysis into an interactive management view. It brings together financial KPIs, vendor contribution, brand sales, ABC analysis and low-performing vendor indicators.



Dashboard KPIs: \$451.62M Sales | \$321.90M Purchase | \$129.72M Gross Profit | 28.7% Profit Margin | \$15.60M Unsold Capital

Operational visuals: Purchase Contribution %, Brand Sales Pareto & ABC Analysis, Top Brands by Sales, Top Vendors by Sales, Low Performing Vendors, A-Tier Items and Items Driving 70% of Sales.

This final layer translates the analytical work into a format that a procurement, category-management or business stakeholder can use for exploration and decision-making.

18. Final Recommendations

- **Manage supplier concentration.** Maintain strategic relationships with the largest vendors while qualifying alternatives for critical categories.
- **Prioritize A-tier products.** The 7% of items driving 70% of sales deserve tighter replenishment and forecasting controls.
- **Attack unsold capital.** Use value, turnover and demand segmentation to prioritize the \$15.60M tied up in unsold inventory.
- **Use bulk purchasing selectively.** Capture volume discounts where demand and turnover justify larger orders.
- **Investigate freight-cost outliers.** Vineyard Brands LLC and Southern Glaziers W&S of NE show freight costs 90–100× the portfolio norm relative to purchase value — worth a direct logistics/invoice review.
- **Differentiate vendor strategy by breadth vs. depth.** High-SKU-count vendors (e.g. Martignetti) and high-revenue-per-SKU vendors (e.g. Diageo) call for different negotiation and replenishment approaches — a scorecard should capture both.
- **Grow high-margin/low-sales brands.** Test targeted promotions, distribution improvements and price optimization before considering rationalization.
- **Build a multi-dimensional vendor scorecard.** Combine sales, purchase value, gross profit, margin, stock turnover, SKU efficiency and unsold capital.
- **Separate commercial strategy from inventory strategy.** High-margin products may need demand generation, while high-volume products may need cost optimization.
- **Use the dashboard as a recurring management tool.** Refresh the KPIs periodically to identify changes in vendor concentration, inventory exposure and product performance.

19. Conclusion

This project set out to answer a practical question: across a portfolio of 128 vendors, 9,651 products and 10,692 transaction records, where is the business winning, where is capital being wasted, and which decisions would move the numbers? The analysis moved through data preparation in SQL, statistical validation in Python, and interactive reporting in Power BI — and at each stage, the findings converged on the same underlying theme: this business performs well in aggregate (\$451.62M in sales, a healthy 28.7% margin), but that aggregate number conceals sharply uneven performance underneath it.

Concentration shows up everywhere once you look for it. Just 10 of 128 vendors account for 65.69% of purchasing. Just 7% of brands — 673 A-tier items — drive 70% of all sales. And within that same portfolio, two vendors pursue completely different playbooks to get there: Diageo North America wins through a small, high-performing catalog (5.9× more revenue per SKU than the portfolio's broadest-catalog vendor), while Martignetti Companies wins through sheer breadth. Neither approach is wrong, but a single "top vendor" metric would flatten that distinction and lead to the wrong follow-up questions.

The margin data tells a similarly counter-intuitive story. Low-performing vendors by sales actually carry meaningfully *higher* margins (41.57% vs. 31.18%), a difference confirmed statistically significant ($T = -17.67$, $p < 0.0001$) rather than left as a coincidence. That single result reframes what "low-performing" should mean in a vendor scorecard — these are not failing vendors, they are reach-constrained vendors with room to grow. The same discipline applied to individual brands surfaced 198 specific low-sales, high-margin products worth a promotional push rather than a delisting.

Not every finding was about upside. \$15.60M is currently tied up in unsold inventory, concentrated in a identifiable set of low-turnover vendors. Two vendors carry a freight cost 90–100× the portfolio norm relative to what they ship, a concrete anomaly that would have stayed invisible inside a per-vendor sales ranking. Both are the kind of finding that only surfaces when a metric is checked at the right grain rather than assumed to behave consistently across the dataset.

Central conclusions

#	Finding
1	Supplier and product activity is highly concentrated — the top 10 vendors and top 7% of brands account for the large majority of commercial activity.
2	A small product group drives a disproportionate share of sales, meaning inventory and service-level policy should follow business impact, not SKU count.
3	Significant capital (\$15.60M) is tied up in unsold inventory, concentrated among a specific, identifiable group of low-turnover vendors.
4	Bulk purchasing can materially reduce unit purchase cost (up to 72% at large order volumes) when used selectively and matched to real demand.
5	Vendor performance cannot be evaluated using margin or sales alone — the two metrics move independently, and sometimes in opposite directions, across the vendor base.
6	Vendors pursue distinct and equally valid strategies around catalog breadth versus revenue concentration, which a one-dimensional "top vendor" ranking obscures.
7	Logistics cost is structurally consistent (~0.52% of purchase value) across the vendor base except for two flagged outliers, isolating a specific, actionable follow-up rather than a portfolio-wide problem.

The strongest management opportunity is therefore not any single fix, but a balanced operating model built around supplier resilience, product prioritization, inventory efficiency, pricing discipline and profitability — applied through a vendor scorecard that captures sales, margin, turnover, SKU efficiency and unsold capital together, rather than any one number in isolation.

Skills & techniques demonstrated

Area	Applied in this project
SQL	CTE-based joins across purchases, sales and freight tables; aggregation at the correct grain to avoid fan-out and double-counting.
Python / pandas	Data cleaning, outlier and negative-value handling, feature engineering (profit margin, stock turnover, HasSales flag), groupby-based segmentation.
Statistics	95% confidence intervals via t-distribution; independent-samples hypothesis testing (T-test) to validate margin differences between vendor groups.
Data visualization	Distribution analysis, correlation heatmaps, Pareto/ABC charts, scatter-based segmentation — all built in Python (matplotlib/seaborn).
Power BI / DAX	Weighted-average measures, percentile-based segmentation (TargetBrand flag), multi-page interactive dashboarding.
Business framing	ABC/Pareto analysis, vendor concentration (HHI-style thinking), unsold capital quantification, procurement-domain recommendations.

Portfolio positioning: This project is designed to demonstrate end-to-end Data Analyst capability — from raw transactional data through statistical validation to an executive-ready dashboard — while remaining directly relevant to procurement and supply-chain decision making.

All charts and figures in this report were generated directly from the project's own SQL/Python pipeline and Power BI dashboard.

Appendix — Key Portfolio Metrics

Area	Finding
Financial performance	\$451.62M sales; \$321.90M purchases; \$129.72M gross profit; 28.7% margin
Unsold capital	\$15.60M on final Power BI dashboard
Vendor concentration	Top 10 vendors = 65.69% of purchases
ABC analysis	673 A-tier items; 7% of items drive first 70% of sales
Low-sales/high-margin brands	198 brands identified in the analysis
Bulk purchasing	Large orders average \$10.78 unit purchase price vs \$39.07 for small orders
Portfolio breadth vs. concentration	Diageo North America: 5.9× more revenue per SKU than Martignetti Companies (396 vs. 1,388 SKUs)
Freight cost efficiency	~0.52% of purchase value portfolio-wide; 2 vendors flagged at 90–100× that rate
Margin comparison	Top vendors mean 31.18%; low vendors mean 41.57%
Statistical test	T = -17.67, p < 0.0001 — null hypothesis rejected

Next deliverables: GitHub README → LinkedIn project entry → LinkedIn launch post → resume bullets → interview walkthrough.